

Space Charge Behaviors of PP/POE/ZnO Nanocomposites for HVDC Cables

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ABSTRACT

Polypropylene (PP) has been paid much attention due to its high melting point, excellent electrical insulating performance and thermoplastic property, which is potential to replace the XLPE as HVDC cable insulating material. Blending PP with polyolefin elastomer (POE) is an effective way to modify its stiffness and brittleness at room temperature. However, space charge behavior of PP/POE blend, as a great concern under dc stress, is not clear and needs further investigation. To research the space charge behaviors, pure PP, PP/POE blend and its nanocomposites with different contents were prepared. Then, mechanical properties, permittivity constant, breakdown strength, volume resistivity, space charge behaviors and trap level distribution were investigated. The results indicate the addition of POE enhances the mechanical flexibility of PP greatly, and nano-sized ZnO doping has little effect on the mechanical flexibility of PP/POE blend. The nanocomposites show lower permittivity constant, higher breakdown strength and higher volume resistivity than the PP/POE blend. Compared to pure PP, the space charge accumulation and electric field distortion get severe in PP/POE blend. However, by nanoparticles doping, space charges are remarkably suppressed, which is related to the increased trap level density in nanocomposites. It indicates the PP/POE/ZnO nanocomposites have much potential for HVDC cable application, which show high mechanical flexibility as well as excellent electrical performance.

Index Terms - HVDC cable, polypropylene, stiffness, brittleness, polyolefin elastomer, insulation, ZnO, space charge, trap level density.

1 INTRODUCTION

DUE to the excellent mechanical and electrical properties, the cross-linked polyethylene (XLPE) is widely used as insulating material in high voltage cables. However, due to its thermosetting characteristic, the recycle of XLPE is difficult at the end of lifetime. Besides, the by-products during crosslinking process can cause space charge formation in the bulk, which is a major obstacle for the application of XLPE in HVDC cables [1-4]. Therefore, it is necessary to investigate new recyclable and reliable insulating materials for DC cables.

Polypropylene (PP) with excellent electrical insulating performance and high melting point has been paid much attention in the literature, which is potential to replace the XLPE as DC cable insulating materials [5-8]. As thermoplastic material, PP is easy to be recycled. Besides, it

can be manufactured without the crosslinking step, reducing much time and energy consumption. However, the poor flexibility of PP makes it impossible to be applied as cable insulating material directly.

Many researches have been focusing on modifying the mechanical performance of PP. It was revealed that the most effective and convenient method of PP toughening was blending in elastomer such as POE, EPDM and LLDPE. Kurahashi et al proved that the low temperature brittleness of PP could be improved by the introduction of ethylene-octene co-polymer (a kind of POE) and LLDPE [9]. Ma et al reported that the introduction of POE in PP significantly increased the notched impact toughness, while the rigidity and tensile strength were reduced [10]. Green et al explored the dynamic mechanical properties and electrical breakdown strength of a propylene homopolymer/propylene-ethylene copolymer blend and the effect of its composition [11]. Zhou et al investigated the influence of POE content on the mechanical, thermal and electrical properties of PP/POE blend [12]. PP/POE blend is the most representative example of PP toughening. Most

researchers investigated this blend focusing on the morphological, rheological and mechanical aspects [5, 13]. For application in DC cables, the electrical properties, especially space charge behavior, are of great significance [12, 14, 15]. However, the space charge characteristics are not completely clear, and more work needs to be done to have a thorough understanding.

In the operation of DC cables, space charge accumulation is a great concern, which can cause electric field distortion and increase the local field within the insulating materials, leading to partial discharge, faster degradation and even premature breakdown [16-21]. Many studies indicated that nanoparticles doping showed great performance in suppressing space charge accumulation in dielectric materials [22-25]. Dakka et al observed that the quantity of space charges was remarkably reduced by adding a small amount of nanoparticles to the polypropylene [26]. Montanari et al investigated the electrical properties of PP filled by nano-sized layered silicates and achieved an improved space charge behavior [27]. It was reported the extensive interface regions between nanoparticles and neighboring materials were closely related to the space charge suppression [28-32]. Generally, less accumulation of space charge would benefit the safe operation of power equipment. Therefore, it is critical to investigate the space charge behavior of PP/POE blend and its nanocomposites to develop new insulating materials for HVDC cables.

In this paper, pure PP, PP/POE blend and PP/POE/ZnO nanocomposites with different contents were prepared. Meanwhile, mechanical properties, permittivity constant, breakdown strength, volume resistivity, space charge behaviors and trap level distribution were investigated. It is observed that the PP/POE blend and its nanocomposites show enhanced mechanical flexibility. In addition, the introduction of POE in PP leads to severe space charge accumulation while nanoparticles doping reduces the quantity of space charges remarkably, which is related to the increased trap level density. The results indicate the PP/POE/ZnO nanocomposites have much potential for HVDC cable application, which show high mechanical flexibility as well as excellent electrical performance.

2 EXPERIMENTAL ARRANGEMENT

2.1 SAMPLE PREPARATION

The PP was supplied by SINOPEC Yangzi petrochemical Co., Ltd., China, the isotacticity of which is about 97 %. The POE (supplied by DuPont Dow) is a kind of ethylene-octene copolymer. Nano ZnO particles used in this study were supplied by Nanjing Haitai Co., Ltd., China with an average particle size of ~ 40 nm. Nano particles were dried in a vacuum oven at 180 °C for 24 h before preparation. The preparation process is as the following. Firstly, appropriate quantities of nanoparticles were mixed uniformly with POE in a two roll mill set to prepare master batch with a rotation speed of 50 rpm for 10 min at the temperature of 130 °C. Secondly, PP was mixed with appropriate quantities of master

batch or pure POE for 10 min at the temperature of 190 °C. Finally, the mixture was molded in a stainless steel mold at the temperature of 190 °C for 10 min and cooled to room temperature under a pressure of 15 MPa. For the convenience of description, the designation, composition and weight ratio of each type of specimen are given in Table 1.

Table 1. Each type of the specimen studied.

Designation	PP (wt%)	POE (wt%)	ZnO(wt%)
P100	100	0	0
P60E40	60	40	0
P60E40N0.5	60	40	0.5
P60E40N2	60	40	2

2.2 MECHANICAL TEST

Tensile test was performed according to ASTM D638 at a temperature of 25 °C with the stretching rate of 30 mm/min. The shape of specimen tested and a typical tensile stress-strain curve are shown in Figure 1. Dog-bone shaped specimen is used for the mechanical tests, with the gauge length of 16 mm, the thickness of 1 mm, and the width of 4 mm. The final mechanical properties are evaluated by the tensile strength and breaking elongation, which is the indicator of rigidity and toughness, respectively. The tensile strength is the quotient of the highest tensile force divided by sectional area, and the breaking elongation is the quotient of the increased length when broken divided by the gauge length. Six specimens were tested for each group and the average value was recorded.

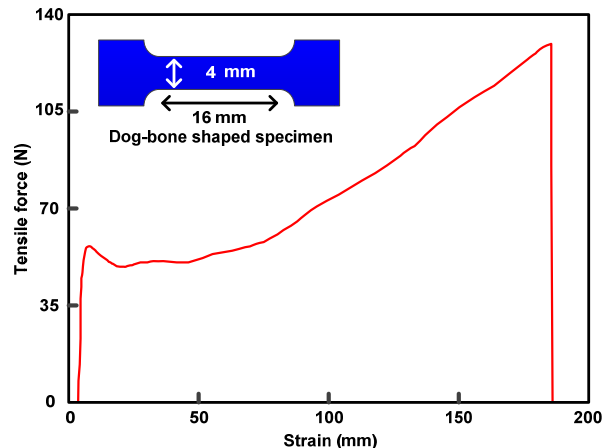


Figure 1. A typical tensile stress-strain curve.

2.3 SPACE CHARGE MEASUREMENT

Space charge measurements were performed by PEA method at room temperature [33]. The experiment arrangement is shown in Figure 2. The semi-conductive material was used as anode (the upper electrode) and aluminum was used as cathode (the lower electrode). Each specimen with a thickness of 250 μ m was stressed under a positive DC voltage with the electric field of 70 kV/mm. The DC poling for all specimens was performed for 3600 s, and the measurement continued for 600 s after poling. The space charge behavior of each group was confirmed by repeating the

tests five times. All the specimens were short-circuited and thermally treated at 80 °C for 48 h and then cooled to room temperature in a vacuum oven before each measurement.

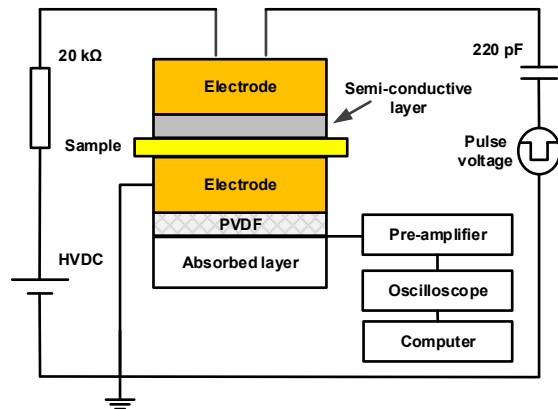


Figure 2. Schematic illustration of the experimental setup for PEA.

2.4 TRAP LEVEL DISTRIBUTION MEASUREMENT

The trap level distributions in PP/POE blend and its nanocomposites were measured with the modified isothermal discharge current (MIDC) test, according to Lei's theory [34]. Tian et al successfully measured the trap level distributions of LDPE and its nanocomposites with this method [35, 36]. In the MIDC test, charges were firstly injected into the specimen under -50 kV/mm for 60 min at 50 °C. Then at the same temperature, the specimen was short-circuited to record the discharge current caused by detrapped electrons.

In the case of a continual trap level distribution with the function $N_t(E)$, which is spatially uniform deep into the sample with a distance l . The induced current in the external circuit during IDC measurement can be approximatively calculated as follows [34]:

$$I = \frac{el^2 kT}{2dt} f_0(E) N_t(E) \quad (1)$$

where e is the electronic charge quantity. k is the Boltzmann constant. t is the measurement time. d is the thickness of the film. E is the trap level. $f_0(E)$ is the initial occupancy of a trap level and approximately equal to 1 based on the assumption that all the traps are fulfilled in this experiment.

The energy gap can be defined as [37, 38]:

$$E_t = kT \ln(\nu t) \quad (2)$$

where E_t is the trap depth. T is the absolute temperature, which is 323.15 K in this paper. ν is the escape frequency of trapped electrons, which can be evaluated from thermally stimulated discharge current (TSDC) measurement, and the typical value of ν is in the range of $10^{11} \sim 10^{13} \text{ (s}^{-1}\text{)}$ [39]. In this paper, an approximate value of 10^{12} s^{-1} was adopted [34, 40]

Then the trap level density distribution can be calculated based on equations (1) and (2):

$$\frac{N_t(E)}{E_t} = \frac{It}{\frac{el^2 kT}{2d} f_0(E) kT \ln(\nu t)} \quad (3)$$

3 RESULTS AND DISCUSSION

3.1 SAMPLE CHARACTERISTICS

In order to investigate the phase structure of PP/POE blend, the specimen fractured surface was etched by heptane in ultrasonic condition for 20 min and then washed with alcohol before Scanning Electron Microscope (SEM) test. The dispersion of POE in the blend can be observed in Figures 3a and 3b. The dark holes represent POE particles, which were removed by heptane etching. POE particles disperse uniformly in the PP matrix with an average diameter of 1 μm , forming a typical sea-island structure. Figures 3c and 3d show the distribution of nano-ZnO particles in the PP/POE bulk. There is almost no agglomeration in the polymer matrix, and the nano-ZnO particles disperse separately. It can be seen the substance consists of two phases, and the smooth one is POE, which appears as segregated phase in the continuous PP matrix. Most nano-ZnO fillers seem to disperse in the PP phase, and less isolated particles appear in the POE matrix. In ternary composites, several factors influence the localization of nanoparticles, such as polarity, melt viscosity, wetting parameter and blending sequence [41]. Inappropriate sequence may lead to severe agglomeration of nanoparticles [10, 42]. In this work, the blending sequence adopted results in separately dispersed nanoparticles. Besides, it indicates obviously that most ZnO nanoparticles migrated to the PP phase from POE phase during the preparation process, which reveals that the nano-ZnO particles have a stronger affinity for PP than for the POE [43].

Mechanical property is an important factor for cable insulation application. In order to characterize the mechanical properties, tensile properties for pure PP, PP/POE and its nanocomposites have been obtained. The values of breaking elongation are shown in Figure 4a. Due to the rubber elastic properties of POE, the breaking elongation of PP/POE is greatly increased compared to the pure PP, which indicates high mechanical flexibility for cable insulation application. Though the breaking elongation decreases a little with the addition of nano-ZnO, it is still much higher than that of pure PP. Figure 4b shows the tensile strength of different specimens, which is an indicator of rigidity. Due to the low tensile strength of POE, the tensile strength of PP/POE blend is lower than that of PP. With the addition of nano-sized ZnO, the tensile strength is increased due to the formation of interaction between nano-ZnO and the polymer chain, which compensates the decrease of toughness in nanocomposites. It can be seen that the PP/POE/ZnO nanocomposites shows high mechanical flexibility and a relatively higher tensile strength than PP/POE blend.

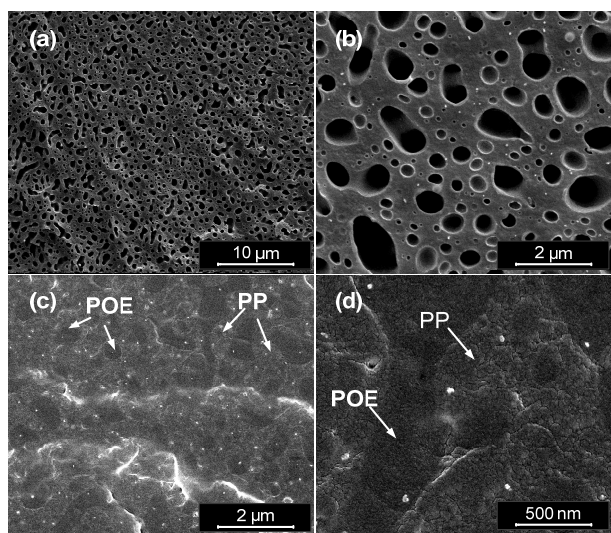


Figure 3. SEM pictures of the specimen fracture surface.

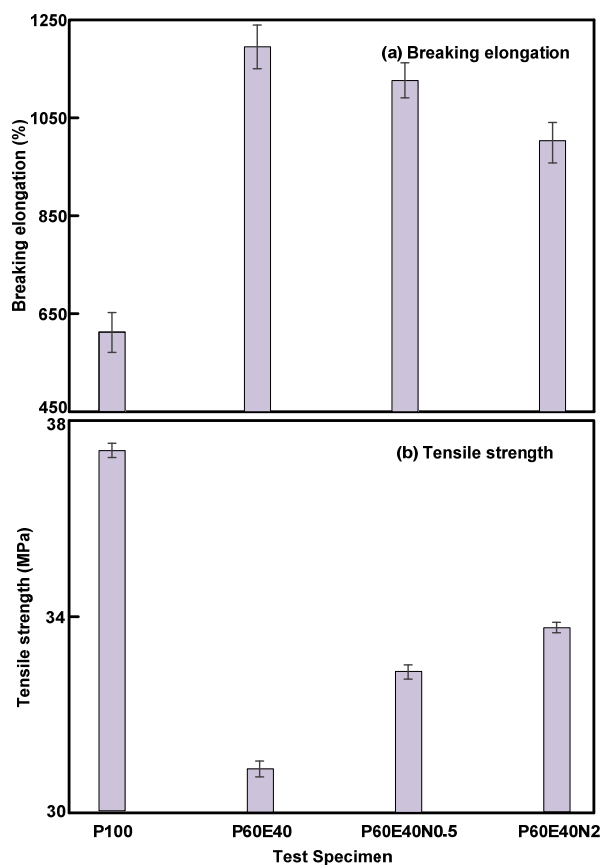


Figure 4. Mechanical characteristics of different specimens.

Figure 5 shows the relationship between the relative permittivity and the frequency. It can be seen that though nano-ZnO particles with higher permittivity are introduced, the relative permittivity of the nanocomposites is lower than that of PP/POE blend. This is consistent with the earlier results in the literature that the permittivity values of nanocomposites were lower than that of the unfilled polymer [44-46]. It is ascribed to the hindrance in the movement of polymer chains, which is possibly due to the entanglements of the polymer chains and the

interaction of the polymer with the nanoparticles [32,47,48]. As a result, the bulk polarization in PP/POE blend is restricted, and the nanocomposites show lower relative permittivity than PP/POE blend. As non-polar material, the permittivity of pure PP and PP/POE blend is almost constant with frequency. However, the permittivity of nanocomposites shows a slight increase at the low frequencies, which is probably caused by interfacial polarization between nano-ZnO and the polymer matrix [49].

The Weibull plots of dc breakdown strength of different specimens are shown in Figure 6. It indicates the PP/POE blend has a lower breakdown strength than pure PP. It may result from the extensive interfaces between POE and PP phase as shown in Figure 3, which can trap charge carriers and distort electric field. Besides, it was reported that the crystallization temperature and crystallinity of PP/POE were lower than pure PP, therefore, the lamellae thickness and free volume in PP/POE blend changed, resulting a decrease in dc breakdown strength [12]. It also shows that nano-ZnO doping enhances the dc breakdown strength. The literature has suggested that the enhancement of breakdown strength in nanocomposite is closely related to the space charge and traps in the interfacial regions [29, 50]. It will be confirmed later in Figures 8 and 10 that the incorporation of ZnO particles can suppress space charge accumulation and inhibit local electrical field distortion. Moreover, as it will be shown in Figure 13, the nanocomposites show high trap level density, which can reduce charge mobility and the amount of injected charges. All these factors together give rise to an increase in breakdown strength of PP/POE/ZnO nanocomposites.

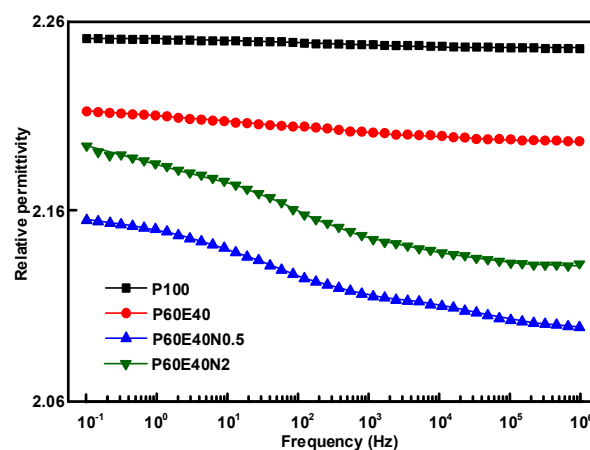


Figure 5. Relationship between the relative permittivity and the frequency.

The volume resistivity was also measured to evaluate the insulating properties as well as the dc breakdown strength [51, 52]. The charging currents of the specimens under 30 kV/mm dc electric field are shown in Figure 7a, which all decay quickly at the initial period and then reach the stable stage. The current values are averaged over the period from 750 to 800 s respectively to calculate the volume resistivity, and the results are shown in Figure 7b. The volume resistivity of PP/POE blend is higher than that of pure PP. This may attribute to the numerous interfaces between POE and PP, which can capture charge carriers, thus resulting in a higher volume resistivity than that of pure PP [12]. It has been reported that due to the restriction of polymer chain

movement in nanocomposites, ions are limited to move over a much shorter range than in undoped specimens, and thus the ion transport are weakened by nanofillers [35, 53]. Furthermore, nano-ZnO particles can introduce traps to reduce charging current. Therefore, the volume resistivity of nanocomposites is higher than that of PP/POE blend.

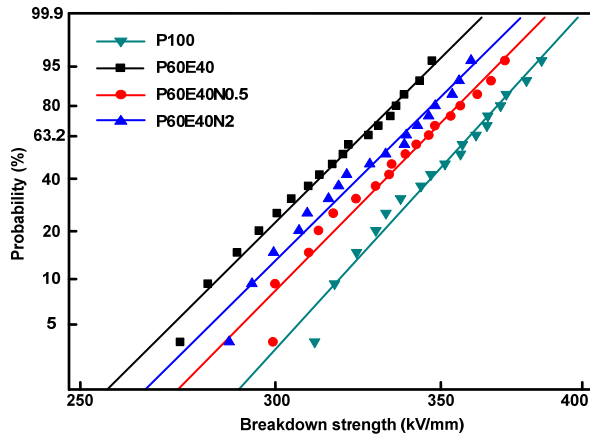


Figure 6. DC breakdown strength of different specimens.

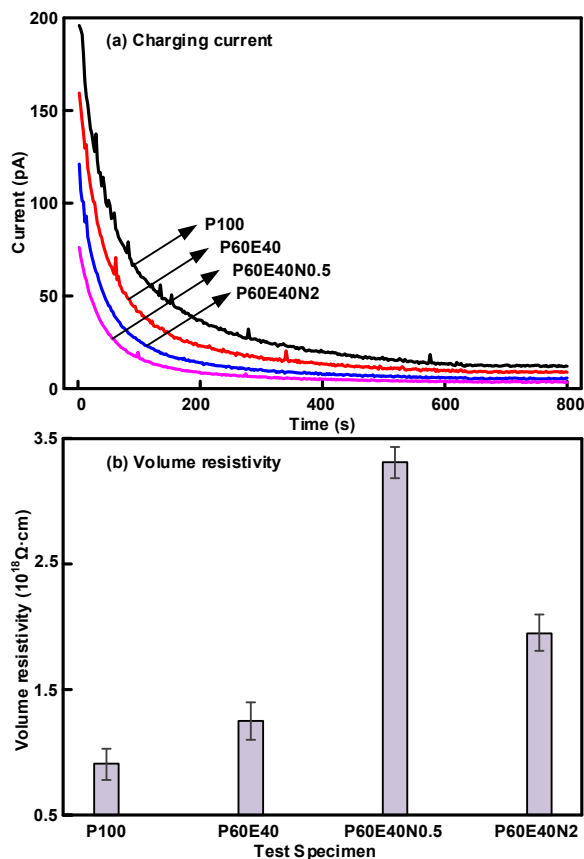


Figure 7. (a) The charging current and (b) volume resistivity of different specimens.

3.2 SPACE CHARGE CHARACTERISTICS

Figure 8 shows the dynamic space charge distributions of different specimens under an dc electric field of 70 kV/mm

for 3600 s. In order to represent the results clearly, space charge distributions in the thickness range of 35 to 215 μm are also shown in the partly-enlarged graphs. For referencing purpose, Figure 8a shows the space charge behavior of pure PP. Due to the semicrystalline nature of PP, the interfaces between the crystalline and amorphous phases can introduce trapping sites, which would capture the injected charges [54].

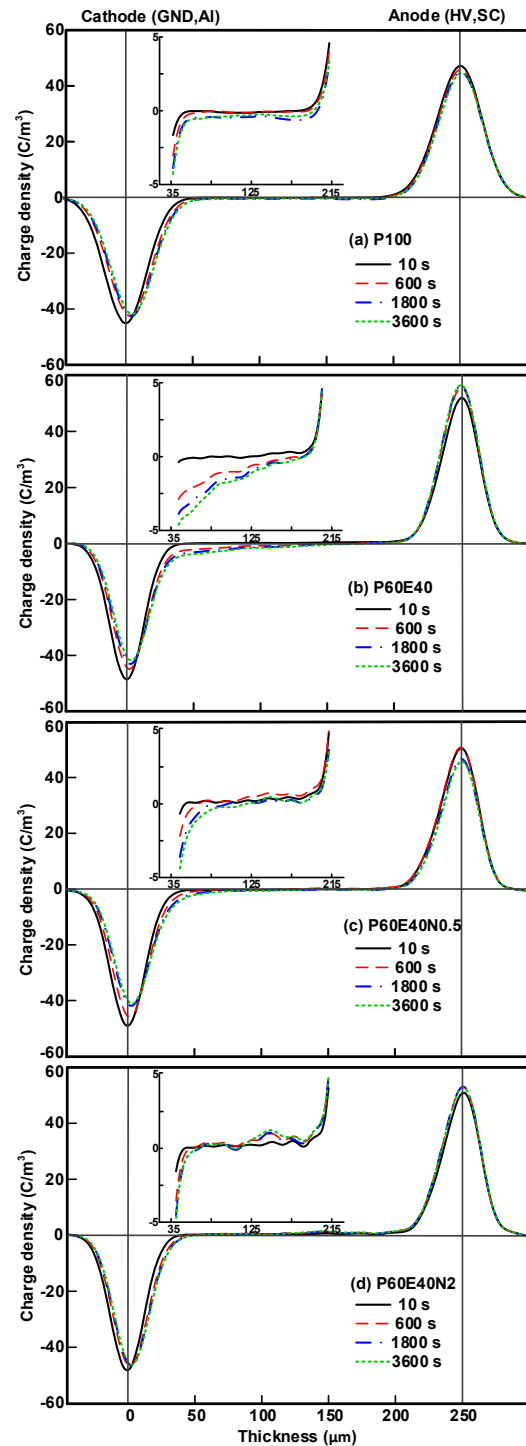


Figure 8. Effect of the poling time at 70 kV/mm DC field on space charge distribution.

With increasing the poling time, homocharges injected from the cathode increase, and heterocharges also appear in the vicinity of anode. It is obvious that the space charge development in the pure PP bulk is minimal.

By comparing the space charge accumulation results in Figures 8a and 8b, it is found that much more charges are injected from cathode in the PP/POE blend. With the lapse of time, the injected charges increase and advance deeper into the middle of the specimen, which is related to the microstructure of the PP/POE blend. With the addition of POE, the interfaces between POE and PP introduce many trapping sites, which can easily capture injected charges during the polarization process [12].

Just by loading the PP/POE blend with a small amount of nano-ZnO, the space charge behavior is improved, as shown in Figures 8c and 8d. The charge accumulation in the vicinity of cathode is obviously suppressed, and the injected charges move much more slowly compared to the PP/POE blend and accumulate in the vicinity of electrodes. Obvious positive charge injection from anode can be seen in the case of P60E40N2, with low charge density and deep injection distance of about 125 μm . On the whole, the space charge behaviors of nanocomposites are dramatically improved with the addition of nano-ZnO compared to that of PP/POE blend.

The total absolute charge amount for each specimen was calculated according to the following equation:

$$Q(t) = \int_0^L |\rho(x, t)| S dx \quad (4)$$

Where $\rho(x, t)$ is the charge density in depth x (distance from the cathode) at poling time t . S is the electrode area. L is the thickness of the specimen. The calculated total charge of each specimen is shown in Figure 9. With the same applied stress, the total charge of pure PP shows the highest start point and is relatively stable. For PP/POE blend, the total charge amount is the highest and keeps increasing with poling time. The total charge amounts of PP/POE/ZnO nanocomposites increase at the initial period and then become stable with poling time. The total charge amount of different specimens after poling for 1 h is in this order: P60E40 > P100 > P60E40N2 > P60E40N0.5.

As the electric field distortion due to space charge formation under DC stress is a great concern, the electric field along the specimen was calculated according to the following equation:

$$E(x) = \frac{1}{\varepsilon_0 \varepsilon_r} \int_0^L \rho(x) dx \quad (5)$$

Where $\rho(x)$ is the charge density, ε_0 is the vacuum permittivity, ε_r is the relative permittivity of the specimen.

The electric field distortion results at the polarization time 3600 s are shown in Figure 10. It can be seen that the electric field is greatly distorted in PP/POE blend bulk compared to that of pure PP. For PP/POE blend, the electric field is inhibited in the vicinity of cathode, while it is enhanced in the vicinity of anode, which is attributed to the large number of

homocharges injected from cathode. Compared to the electric field distribution of PP/POE blend, electric field distortion in 0.5 wt% and 2 wt% nanocomposites gets better. It is observed that the electric field of P60E40N0.5 is in close proximity to the ideal electric field of 70 kV/mm, indicating that the electric field distribution is rather uniform in the polymer. In the case of 2 wt% nanocomposites, the electric field in the vicinity of cathode gets enhanced, which is related to the homocharges injection from anode as shown in Figure 8d.

The distortion factor of the electric field ΔE can be obtained by the following equation:

$$\Delta E = \frac{E_{\max} - E_{av}}{E_{av}} \quad (6)$$

Where the $E_{\max}$ is the highest value of the electric field in the specimen bulk after DC poling of 1 h. E_{av} is the ideal electric field in the absence of space charge. The results are shown in Figure 11. It is observed that the PP/POE blend shows the highest distortion factor, which is increased by 4 times as compared with pure PP. The severe electric field distortion induced by space charge formation may lead to partial discharge in the micro defects, thus accelerating the insulation aging. Nanoparticles doping can greatly mitigate

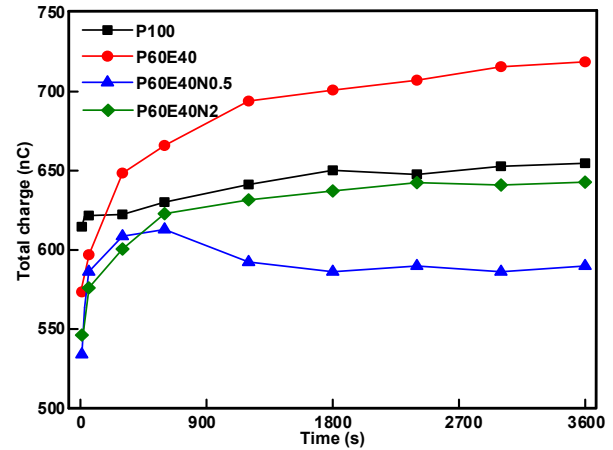


Figure 9. Relationship between the total charge and the poling time.

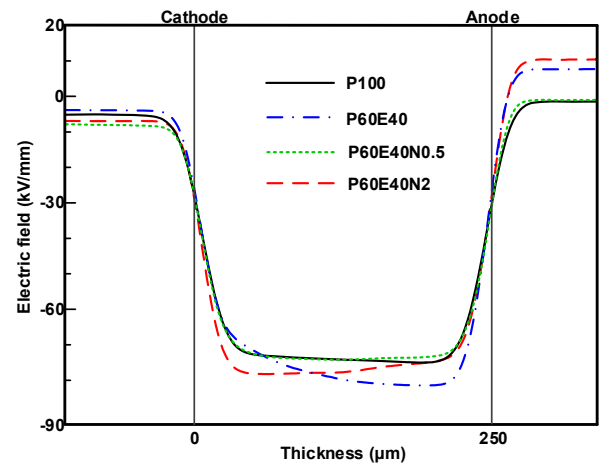


Figure 10. Electric field distributions after 1 h of DC poling.

the electrical field distortion. The distortion factor of specimen with 0.5 wt% nanoparticles is much lower than those of the other specimens, indicating uniform electric field in the polymer.

The comparison of total charge quantity during the depolarization process is shown in Figure 12. Clearly, the quantity of space charges decays rapidly at the initial period and keeps invariant after about one minute in all specimens. It can be seen that after depolarization for 150 s, the quantity of space charges in pure PP bulk is the least, while large number of space charges still remain in the other specimens. Besides, space charge in nanocomposites decays slower than that in the PP/POE blend.

Compared to the results of pure PP, the addition of POE into polypropylene matrix results in more space charge accumulation in the vicinity of cathode and higher electric field distortion factor, which is unexpected for insulating materials in HVDC cable application. While all the PP/POE/ZnO nanocomposites show lower quantity of space charges and lower distortion factor than those of PP/POE blend. It is proved that the nano-ZnO doping can effectively suppress the space charge accumulation and electric field distortion in polymer, which makes the nanocomposites more suitable for HVDC cable application than PP/POE blend.

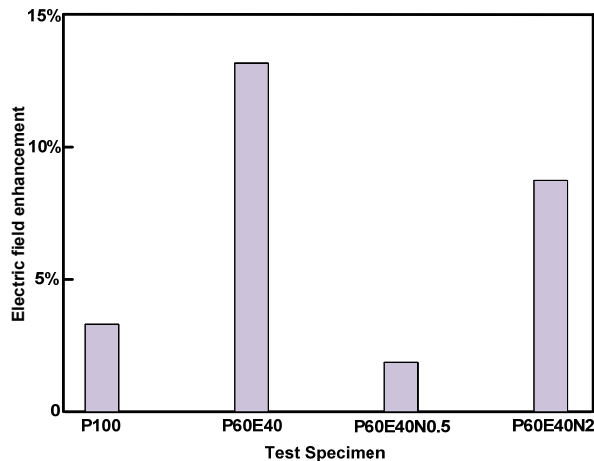


Figure 11. Electric field enhancements after 1 h of DC poling.

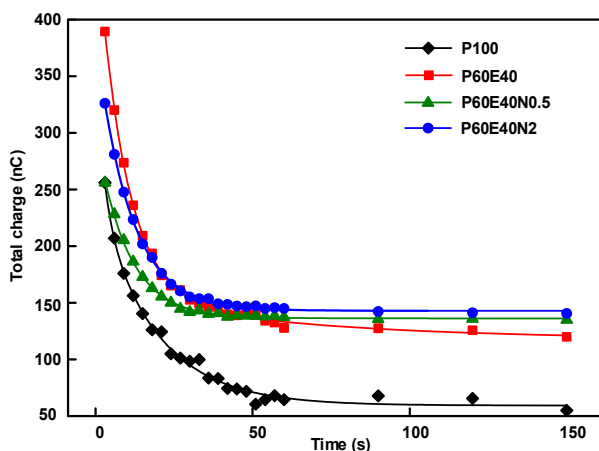


Figure 12. Relationship between total charge and the depolarization time.

3.3 TRAP LEVEL DISTRIBUTION

In order to investigate the influence of nano-sized ZnO on the space charge characteristics furtherly, the trap level distributions of different specimens calculated by equation (3) are shown in Figure 13. The trap levels are all in the range of 0.75 to 1.15 eV, and the peaks locate at ~ 1.03 eV. Due to the trapping sites introduced by the extensive interfaces between POE and PP, the trap level density of PP/POE blend is much higher than that of PP [12, 15]. With the introduction of nano-sized ZnO, the deep trap level density increases, and the trap density peak of P60E40N0.5 is about 3 times that of PP/POE blend.

The results of trap level distribution clearly indicate that nano-ZnO particles introduce large number of deep trap states into the specimens. Due to the large specific surface area of nanoparticles, extensive interface regions are formed on the boundary between nano-ZnO and the polymer matrix. In interface regions, polymers chains are strongly bound and/or interacted to the nanoparticles, which results in partly immobility of the polymer chains together with the space steric hindrance of the nanoparticles. Therefore, the local properties (chain arrangement, free volume, mobility, etc.) can be changed, which may lead to the increase in the trap level density [28-32].

The space charge reduction is related to the increased trap density in nanocomposites. The introduced deep trapping sites can easily capture a number of charges. These localized charges in nanocomposites are less mobile, which is confirmed by the slow transport of injected charges shown in Figure 8 and the slow decay rate of space charge in Figure 12. Moreover, the firstly trapped homocharges accumulate in the vicinity of electrodes, which can result in an independent electric field and decrease the effective electric field between the electrodes and the surface of the specimens, thus forming a blocking layer for further charge injection and suppressing the space charge formation in the specimen bulk. The fact that the nanoparticles doping can result in the increase of trap level density and the suppression of the space charge formation have been confirmed by extensive researches in the literature [35, 55, 56].

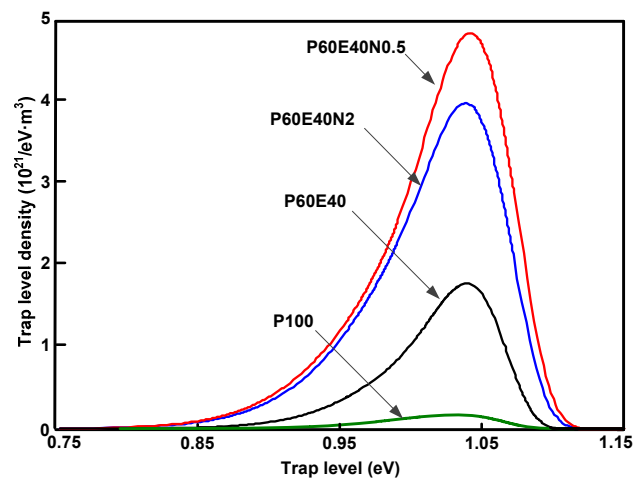


Figure 13. Trap level distribution of different specimens.

4 CONCLUSION

An experimental study of space charge behavior in pure PP, PP/POE blend and its nanocomposites was conducted. The results reveal that the space charge accumulation is obviously influenced by the addition of POE and nano-ZnO particles. The main conclusions can be summarized as follows:

1. Due to the rubber elastic properties of POE, the breaking elongation of PP/POE blend is greatly improved while the tensile strength is reduced. The introduction of nano ZnO into PP/POE blend results in a slight decrease of the breaking elongation but a slight enhancement of the tensile strength. The nanocomposites show high mechanical flexibility as well as the PP/POE blend and a relatively higher tensile strength than PP/POE blend.

2. The addition of POE into PP matrix results in more space charge accumulation and severe electric field distortion, which is due to the interface between POE and PP playing a role of trapping site. Though the PP/POE blend show improved mechanical flexibility, the severe space charge accumulation indicates that it is not suitable to be applied for HVDC cable directly.

3. The introduction of nano-ZnO into PP/POE blend increases the trap level density, which can lead to the reduction in charge injection, thus suppressing space charge accumulation and electric field distortion in the nanocomposites bulk. Besides, the nanocomposites show great performance in permittivity, breakdown strength and volume resistivity. With the improved mechanical properties and excellent electrical properties, the PP/POE/ZnO nanocomposites have much potential for HVDC cable application.

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REFERENCES

- [1] G. C. Montanari, C. Laurent, G. Teyssedre, A. Campus and U. H. Nilsson, "From LDPE to XLPE: Investigating the Change of Electrical Properties. Part I. Space Charge, Conduction and Lifetime", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 12, No. 3, pp. 438-446, 2005.
- [2] G. Teyssedre, C. Laurent, G. C. Montanari, A. Campus and U. H. Nilsson, "From LDPE to XLPE: Investigating the Change of Electrical Properties. Part II. Luminescence", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 12, No. 3, pp. 447-454, 2005.
- [3] F. N. Lim, R. J. Fleming and R. D. Naybour, "Space Charge Accumulation in Power Cable XLPE Insulation", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 6, No. 3, pp. 273-281, 1999.
- [4] M. Fu, G. Chen, L. A. Dissado and J. C. Fothergill, "Influence of Thermal Treatment And Residues on Space Charge Accumulation in XLPE for DC Power Cable Application", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 14, No. 1, pp. 53-64, 2007.
- [5] I. L. Hosier, S. Reaud, A. S. Vaughan and S. G. Swingle, "Morphology, Thermal, Mechanical and Electrical Properties of Propylene-Based Materials for Cable Applications", *IEEE Int'l. Sympod. Electr. Insul.*, pp. 502-505, 2008.
- [6] M. S. Hamzah, M. Jaafar and M. K. M. Jamil, "Electrical Insulation Characteristics of Alumina, Titania, and Organoclay Nanoparticles Filled PP/EPDM Nanocomposites", *J. Appl. Polym. Sci.*, Vol. 131, No. 23, pp. 41184, 2014.
- [7] M. Abou-Dakka, "Evolution of Space Charges and Conductivity with Dc Aging of Polyethylene-Synthetic and Natural Clay Composites", *J. Nanomater.*, pp. 463748, 2012.
- [8] I. L. Hosier, L. Cozzarini, A. S. Vaughan and S. G. Swingle, "Propylene Based Systems for High Voltage Cable Insulation Applications", *J. Phys.: Conf. Series*, Vol. 183, No. 1, pp. 012015, 2009.
- [9] K. Yoshino, T. Demura, M. Kawahigashi, Y. Miyashita, K. Kurahashi and Y. Matsuda, "The Application of Novel Polypropylene to The Insulation of Electric Power Cable", *IEEE Trans. Distrib. Conf. Exhibit.*, pp. 1278-1283, 2002.
- [10] C. G. Ma, Y. L. Mai, M. Z. Rong, W. H. Ruan and M. Q. Zhang, "Phase Structure and Mechanical Properties of Ternary Polypropylene/Elastomer/Nano-CaCO₃ Composites", *Compos. Sci. Technol.*, Vol. 67, No. 14, pp. 2997-3005, 2007.
- [11] C. Green, A. Vaughan, G. Stevens, A. Pye, S. Sutton, T. Geussens and M. Fairhurst, "Thermoplastic Cable Insulation Comprising A Blend of Isotactic Polypropylene and A Propylene-Ethylene Copolymer", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 22, No. 2, pp. 639-648, 2015.
- [12] Y. Zhou, J. L. He, J. Hu, X. Y. Huang and P. K. Jiang, "Evaluation of Polypropylene/Polyolefin Elastomer Blends for Potential Recyclable HvdC Cable Insulation Applications", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 22, No. 2, pp. 673-681, 2015.
- [13] A. L. N. Silva, M. I. B. Tavares, D. P. Politano, F. M. B. Coutinho and M. C. G. Rocha, "Polymer Blends Based on Polyolefin Elastomer and Polypropylene", *J Appl Polym Sci*, Vol. 66, No. 10, pp. 2005-2014, 1997.
- [14] Y. Zhou, J. L. He, J. Hu and B. Dang, "Surface Modified MgO Nanoparticle Enhances the Mechanical and DC Electrical Characteristics of Polypropylene/Polyolefin Elastomer Nanodielectrics", *J. Appl. Polym. Sci.*, Vol. 133, No. 1, pp. 42863, 2016.
- [15] B. Dang, J. L. He, J. Hu and Yao Zhou, "Large Improvement in Trap Level and Space Charge Distribution of Polypropylene by Enhancing the Crystalline-Amorphous Interfaces Effect in Blends", *Polym. Int.*, Vol. 65, No. 4, pp. 371-379, 2016.
- [16] T. T. N. Vu, G. Teyssedre, B. Vissouvanadin, S. Le Roy and C. Laurent, "Correlating Conductivity and Space Charge Measurements in Multi-dielectrics Under Various Electrical and Thermal Stresses", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 22, No. 1, pp. 117-127, 2015.
- [17] D. Fabiani, G. C. Montanari, A. Cavallini and G. Mazzanti, "Relation Between Space Charge Accumulation and Partial Discharge Activity in Enamelled Wires Under PWM-like Voltage Waveforms", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 11, No. 3, pp. 393-405, 2004.
- [18] G. Mazzanti and G. C. Montanari, "Elemental Strain and Trapped Space Charge in Thermoelectrical Aging of Insulating Materials Life Modeling", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 8, No. 6, pp. 966-971, 2001.
- [19] L. A. Dissado, G. Mazzanti and G. C. Montanari, "The Role of Trapped Space Charge in the Electrical Aging of Insulating Materials", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 4, No. 5, pp. 496-506, 1997.
- [20] G. C. Montanari, "Bringing an Insulation to Failure: The Role of Space Charge", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 18, No. 2, pp. 339-364, 2011.
- [21] M. Fu, L. A. Dissado, G. Chen and J. C. Fothergill, "Space charge formation and its modified electric field under applied voltage reversal and temperature gradient in XLPE cable", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 15, No. 3, pp. 851-860, 2008.
- [22] T. Tanaka, A. Bulinski, J. Castellon, M. Frechette, S. Gubanski, J. Kindersberger, G. C. Montanari, M. Nagao, P. Morshuis, Y. Tanaka, S. Pelissou, A. Vaughan, Y. Ohki, C. W. Reed, S. Sutton and S. J. Han, "Dielectric Properties of XLPE/SiO₂ Nanocomposites Based on CIGRE WG D1.24 Cooperative Test Results", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 18, No. 5, pp. 1484-1517, 2011.
- [23] C. Calebrese, H. Le, L. S. Schadler and J. K. Nelson, "A Review on the Importance of Nanocomposite Processing to Enhance Electrical Insulation", *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 18, No. 4, pp. 938-945, 2011.
- [24] T. Maezawa, J. Taima, Y. Hayase, Y. Tanaka, T. Takada, Y. Sekiguchi and Y. Murata, "Space Charge Formation in LDPE/MgO Nano-

- composite under High Electric Field at High Temperature", IEEE Conf. Electr. Insul. Dielectr. Phenomena, pp.271-274, 2007.
- [25] S. T. Li, G. L. Yin, G. H. Chen, J. Y. Li, S. N. Bai, L. S. Zhong, Y. X. Zhang and Q. Q. Lei, "Short-Term Breakdown and Long-Term Failure in Nanodielectrics: A Review", IEEE Trans. Dielectr. Electr. Insul., Vol. 17, No. 5, pp. 1523-1535, 2010.
- [26] M. A. Dakka, A. Bulinski and S. S. Bamji, "Space Charge Evolution in Polypropylene Containing Synthetic and Natural Organoclayes," IEEE Conf. Electr. Insul. Dielectr. Phenomena, pp.1-4, 2010.
- [27] G. C. Montanari, D. Fabiani and F. Palmieri, "Modification of electrical properties and performance of EVA and PP insulation through nanostructure by organophilic silicates", IEEE Trans. Dielectr. Electr. Insul., Vol. 11, No. 5, pp. 754-762, 2004.
- [28] T. Tanaka, M. Kozako, N. Fuse and Y. Ohki, "Proposal of A Multi-Core Model for Polymer Nanocomposite Dielectrics", IEEE Trans. Dielectr. Electr. Insul., Vol. 12, No. 4, pp. 669-681, 2005.
- [29] T. Takada, Y. Hayase, Y. Tanaka and O. Tatsuki, "Space Charge Trapping in Electrical Potential Well Caused by Permanent and Induced Dipoles for LDPE/Mgo Nanocomposite", IEEE Trans. Dielectr. Electr. Insul., Vol. 15, No. 1, pp. 152-160, 2008.
- [30] T. Tanaka, "Dielectric Nanocomposites with Insulating Properties", IEEE Trans. Dielectr. Electr. Insul., Vol. 12, No. 5, pp. 914-928, 2005.
- [31] J. K. Nelson and J. C. Fothergill, "Internal Charge Behaviour of Nanocomposites, Nanotechnology, Vol. 15, No. 5, pp. 586-595, 2004.
- [32] M. Roy, J. K. Nelson, R. K. MacCrone and L. S. Schadler, "Candidate Mechanisms Controlling the Electrical Characteristics of Silica/XLPE Nanodielectrics", J. Mater. Sci., Vol. 42, No. 11, pp. 3789-3799, 2007.
- [33] Y. Li, M. Yasuda and T. Takada, "Pulsed electroacoustic method for measurement of charge accumulation in solid dielectrics", IEEE Trans. Dielectr. Electr. Insul., Vol. 1, No. 2, pp. 188-195, 1994.
- [34] Q. Q. Lei, F. Q. Tian, C. Yang, L. J. He and Y. Wang, "Modified Isothermal Discharge Current Theory and its Application in the Determination of Trap Level Distribution in Polyimide Films", J. Electrostat., Vol. 68, No. 3, pp. 243-248, 2010.
- [35] F. Q. Tian, J. Y. Yao, P. Li, Y. Wang, M. L. Wu and Q. Q. Lei, "Stepwise Electric Field Induced Charging Current and Its Correlation with Space Charge Formation in LDPE/ZnO Nanocomposite", IEEE Trans. Dielectr. Electr. Insul., Vol. 22, No. 2, pp. 1232-1239, 2015.
- [36] F. Q. Tian, Q. Q. Lei, X. Wang and L. Wang, "Investigation of Electrical Properties of LDPE/ZnO Nanocomposite Dielectrics", IEEE Trans. Dielectr. Electr. Insul., Vol. 19, No. 3, pp. 763-769, 2012.
- [37] H. Sjöstedt, S. M. Gubanski and Y. V. Serdyuk, "Charging Characteristics of EPDM and Silicone Rubbers Deduced from Surface Potential Measurements", IEEE Trans. Dielectr. Electr. Insul., Vol. 16, No. 3, pp. 696-703, 2009.
- [38] P. K. Watson, "The Energy Distribution of Localized States in Polystyrene, Based on Isothermal Discharge Measurements", J. Phys. D: Appl. Phys., Vol. 23, No. 12, pp. 1479-1484, 1990.
- [39] R. Chen and Y. Kirsh, *Analysis of Thermally Stimulated Processes*, Oxford Pergamon Press, 1981.
- [40] F. Yoshida and M. Yoshiura, "Charging Property and Charge Trap Parameters in Porous Polypropylene Film using Thermally Stimulated Current" Proc. IEEE Int. Conf. Solid Dielectr., 2010.
- [41] M. Jarnthong, C. Nakason, N. Lopattananon and Z. Peng, "Influence of Incorporation Sequence of Silica Nanoparticles on Morphology, Crystallization Behavior, Mechanical Properties, and Thermal Resistance of Melt Blended Thermoplastic Natural Rubber", Polym. Compos., Vol. 33, No. 11, pp. 1911-1920, 2012.
- [42] C. G. Martins, N. M. Larocca, D. R. Paul and L. A. Pessan, "Nanocomposites Formed from Polypropylene/EVA Blends", Polymer, Vol. 50, No. 7, pp. 1743-1754, 2009.
- [43] S. H. Lee, M. Kontopoulou and C. B. Park, "Effect of Nanosilica on the Co-Continuous Morphology of Polypropylene/Polyolefin Elastomer Blends", Polymer, Vol. 51, No. 5, pp. 1147-1155, 2010.
- [44] Y. Cao, P. C. Irwin and K. Younsi, "The Future of Nanodielectrics in the Electrical Power Industry", IEEE Trans. Dielectr. Electr. Insul., Vol. 11, No. 5, pp. 797-807, 2004.
- [45] J. K. Nelson and Y. Hu, "Nanocomposite Dielectrics – Properties and Implications", J. Phys. D: Appl. Phys., Vol. 38, No. 2, pp. 213-222, 2005.
- [46] S. Singha and M. J. Thomas, "Dielectric Properties of Epoxy Nanocomposites", IEEE Trans. Dielectr. Electr. Insul., Vol. 15, No. 1, pp. 12-23, 2008.
- [47] S. Singha and M. J. Thomas, "Influence of Filler Loading on Dielectric Properties of Epoxy Zn Nanocomposites", IEEE Trans. Dielectr. Electr. Insul., Vol. 16, No. 2, pp. 531-542, 2009.
- [48] S. T. Li, W. W. Wang, S. H. Yu and H. G. Sun, "Influence of Hydrostatic Pressure on Dielectric Properties of Polyethylene/Aluminum Oxide Nanocomposites", IEEE Trans. Dielectr. Electr. Insul., Vol. 21, No. 2, pp. 519-528, 2014.
- [49] L. Zhang, Y. X. Zhou, M. Huang and Y. C. Sha, "Effect of Nanoparticle Surface Modification on Charge Transport Characteristics in XLPE/SiO₂ Nanocomposites", IEEE Trans. Dielectr. Electr. Insul., Vol. 21, No. 2, pp. 424-433, 2014.
- [50] Y. Murakami, M. Nemoto, S. Okuzumi, S. Masuda, M. Nagao, N. Hozumi and Y. Sekiguchi, "DC Conduction and Electrical Breakdown of Mgo/LDPE Nanocomposite", IEEE Trans. Dielectr. Electr. Insul., Vol. 15, No. 1, pp.33-39, 2008.
- [51] R. N. Hampton, "Some of The Considerations for Materials Operating under High-Voltage, Direct-Current Stresses", IEEE Electr. Insul. Mag., Vol. 24, No. 1, pp. 5-13, 2008.
- [52] T. L. Hanley, R. P. Burford, R. J. Fleming and K. W. Barber, "A General Review of Polymeric Insulation for Use in HVDC Cables", IEEE Electr. Insul. Mag., Vol. 19, No. 1, pp. 13-24, 2003.
- [53] T. Miyamoto and K. Shibayama, "Free-volume Model for Ionic Conductivity in Polymers", J. Appl. Phys. Vol. 44, No. 12, pp. 5372-5376, 1973.
- [54] T. J. Lewis, "Polyethylene under Electrical Stress", IEEE Trans. Dielectr. Electr. Insul., Vol. 9, No. 5, pp. 717-729, 2002.
- [55] S. T. Li, N. Zhao, Y. J. Nie, X. Wang, G. Chen and G. Teyssedre, "Space Charge Characteristics of LDPE Nanocomposite/LDPE Insulation System", IEEE Trans. Dielectr. Electr. Insul., Vol. 22, No. 1, pp. 92-100, 2015.
- [56] L. Zhang, Y. X. Zhou, J. H. Tian, Y. C. Sha, Y. X. Zhang, H. Z. Wu and Y. S. Wang, "Experiment and simulation of space charge suppression in LDPE/MgO nanocomposite under external DC electric field", J. Electrostat., Vol. 72, No. 4, pp. 252-260, 2014.

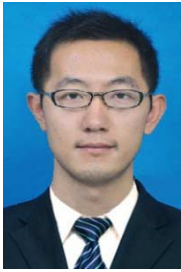


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